

Scale-Aware Conditional Privacy Audits for Graph Neural Network Prediction APIs

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Abstract—We study *conditional membership-privacy audits* for graph ML prediction APIs: leakage depends on structure, architecture, and split—not on “GNNs” or node count alone. SPAB is a scale-aware audit protocol and release artifact reporting Acc, LiRA (primary), TPR@1%FPR, train time, and memory as $(h, \rho, M, n) \mapsto \text{metrics}$, with NeighborLoader systems numbers through ogbn-arxiv ($\sim 169k$). SPAB’s *Volume contribution* is systems cost plus trustworthy negative controls, not proof that large graphs always leak: under the official OGB split, ogbn-arxiv score MIAs sit near chance. Privacy findings concentrate where risk is predicted—Cora GraphSAGE (LiRA $0.567 \rightarrow 0.510$), Citeseer GraphSAGE under val-selected SAMI (LiRA $0.596 \rightarrow 0.538$), and Chameleon (LiRA $0.623 \rightarrow 0.551$)—while Actor compares LBP, and a GCN hard cell / stress test are boundaries, not Volume peers. A three-term LTE heuristic enables risk-weighted ϕ -alignment (SAMI); removing LTE costs ≈ 0.11 Acc. We do not claim a universal leakage law.

Index Terms—graph neural networks, membership inference, privacy audit, large-scale graph analytics, ogbn-arxiv, LiRA

I. INTRODUCTION

A. Motivation

Modern data platforms serve GNN node classifiers [8], [9] behind prediction APIs over citation, payment, and referral graphs. Those APIs create a membership-privacy surface [1]–[4]. For large-scale deployments the right question is not “do GNNs leak?” but *under which structure, architecture, and scale does leakage appear, at what systems cost, and where should a finite privacy budget be spent?*

a) *What we release.*: **SPAB** is a scale-aware *audit protocol* and release CSV (not a claim of community benchmark adoption): LiRA-primary rows with systems fields where measured, through ogbn-arxiv. **SAMI** is one locked-config empirical hardening (LTE heuristic + ϕ -alignment). A synthetic LOO plot is supporting evidence only [5].

b) *Empirical spine.*: Fig. 3 separates *audit regimes* (not a single n -axis claim): near-chance under official ogbn-arxiv and PubMed; elevated LiRA/conf on Actor and Cora; a synthetic cell is a controlled stress test only. **Volume ownership**: scaling NeighborLoader to $\sim 169k$ nodes validates that the audit *runs* and that this protocol finds little score-based membership signal on ogbn-arxiv—it does *not* validate that “large graphs leak.” The privacy mechanism story is Cora GraphSAGE (vs. GTD/LBP/MaskArmor) plus Actor baselines; the GCN hard cell is an architecture boundary.

B. Gap and related defenses

LBP adds global output noise [5]; GTD uses alternate training and loss flattening [18]; MaskArmor truncates confidence tails [19]; GPS targets *attribute* leakage via structure [32]; GAP/NAP-GNN provide ϵ -DP [12], [33]. What is incremental here is not “structure matters,” but a *scalable conditional audit* with systems metrics plus a *risk-budgeted* empirical hardening that allocates effort where local topology predicts membership signal—without claiming DP.

C. Contributions

- 1) **SPAB (primary)**. A scale-aware *audit protocol* and release artifact (not a community benchmark claim): fixed columns, NeighborLoader systems through ogbn-arxiv as negative control, regime map with explicit split tags.
- 2) **One defense (SAMI)**. Locked Cora config + pre-registered Citeseer val-only (λ, σ) ; LiRA wins on Cora, Citeseer, and Chameleon at matched Acc; Actor grid (LBP stronger on LiRA alone); $-LTE$ Acc collapse; honest GCN hard cell.
- 3) **Supporting evidence (not a law)**. Held-out synthetic LOO plot interpreting where score MIAs rise—one figure, demoted from a formal claim.

The remainder of the paper proceeds as follows. Section II reviews related work; Section III the threat model. Section IV states the supporting predictive regularity; Section V presents SAMI. Sections VI–VII evaluate; Section VIII notes boundaries.

II. BACKGROUND AND RELATED WORK

A. Membership inference on classical models

Shokri et al. [1] introduced shadow-model MIAs; Salem et al. [2] showed that simple confidence/entropy features often suffice. Yeom et al. [3] linked attack advantage to overfitting. Carlini et al. [4] argued for likelihood-ratio attacks (LiRA) and low false-positive operating points as security-relevant metrics. Related extraction and inversion attacks [22], [23], [30] further motivate treating prediction APIs as privacy surfaces. We adopt both confidence-based ϕ -attacks (continuity with prior GNN work) and Gaussian LiRA as the primary privacy

TABLE I
PRIOR-ART CONTRAST (SETTING / STRUCTURE / DP / SCALE). GAP
COLUMN IS OUR CONTRIBUTION RELATIVE TO THAT LINE.

Work	Setting	Struct.	DP?	Scale	Our gap
Olatunji+LBP [5]	Score MIA	No	No	Citation	Systems+API protocol
He et al. [35]	Node MIA	Topo.	No	Citation	Audit GAP recipe+DP+GNN
GTD [18]	Transductive	No	No	Citation	Risk-weighted alt. GTD
GPS [32]	Attr. leak	Yes	No	Med.	Membership scores
Khosla [36]	MI risk	Yes	No	Med.	API audit+cost
GAP [12]	DP GNN	Partial	Yes	Large	Emp. matched Acc

TABLE II
METHOD CONTRAST (GUARANTEE TYPE / UTILITY / STRUCTURE / MIA
SCORE COVERAGE).

Method	Guarantee	Utility	Structure	MIA scores
SAMI	Empirical	Low-mod.	Yes (LTE)	Score black-box
ϵ -DP (agg.)	Often high	Partial	Broad via DP	
ϵ -DP (imp.)	Tuned	Yes (imp.)	Broad via DP	
Empirical	High cost	No	Posterior MIA	
Empirical	Low-mod.	No (global)	Loss-gap MIA	

TABLE III
PRIMARY NOTATION.

Symbol	Meaning
n, d, C	Nodes, feature dimension, classes
X, y	Features and labels
$\mathcal{T}, \mathcal{V}, \mathcal{S}$	Train / val / test node sets
$\hat{p}_v, z_v$	Softmax posterior and logits at v
$\phi(v)$	Four-dimensional attack feature map
r_v	LTE leakage risk in $[0, 1]$
$h(G), \rho$	Observed homophily, edge density
ECE	Expected calibration error ($B=10$)

metric, and we report calibration error because overconfident probabilities are a recurring membership cue [15], [31].

B. Membership inference on graphs

Olatunji et al. [5] demonstrated node-level black-box MIAs on GNNs and proposed LBP and neighborhood sampling defenses. He et al. [35] systematically study node-level MIAs under varying topology knowledge. Subsequent work studies graph-level membership [6], embedding leakage [21], and broader attack taxonomies [7]. GTD [18] targets the transductive setting with alternate training and loss flattening. Khosla [36] shows that training-graph construction and inference-time edge access directly modulate membership advantage—orthogonal to our score-API audit protocol. GPS [32] studies structure→attribute leakage; we target score-based *membership*. Our threat model matches the score-based black-box node setting; we do not claim white-box, structural-modification, or federated attack coverage.

C. Architectures, regularization, and privacy

GCNs [8] apply spectral neighborhood averaging; GraphSAGE [9] uses mean aggregation and is often more robust under heterophily. DropEdge [10] randomly deletes edges during training. DP-SGD and DP-GNNs [12], [13] provide formal privacy; SAMI is an empirical mitigation and does **not** claim differential privacy. Homophily modulates the train/test ϕ -gap [16], [17]: under low h and sparsity, supervised memorization widens membership cues—motivating risk-weighted budget allocation.

D. Positioning

Primary claim: **SPAB**—conditional privacy+systems audits for graph ML APIs at scale. SAMI is one optional hardening—not three equal breakthroughs. Table I situates the gap: we do not beat GAP on guarantees; we report deployable LiRA—primary audits with systems cost through ogbn-arxiv and a risk-weighted defense that wins LiRA at matched Acc on Cora, Citeseer (val-selected), and Chameleon under our protocol. We follow modern MIA guidance [4], [34]: defense-aware shadows, TPR@low FPR, label-only gap attacks, and a tuned DP-SGD Pareto anchor.

III. PROBLEM SETUP

A. Notation

Let $G = (V, E)$ be an undirected graph with $n = |V|$ nodes, feature matrix $X \in \mathbb{R}^{n \times d}$, and labels $y \in \{0, \dots, C-1\}^n$. A random split partitions V into supervised train $\mathcal{T}$, validation $\mathcal{V}$, and test $\mathcal{S}$ with ratios 40%/20%/40% (resampled per seed). Training is *transductive*: every node participates in message passing; only $\mathcal{T}$ enters the cross-entropy loss. Table III summarizes notation.

B. Threat model

Definition 1 (Black-box node membership adversary). *The adversary queries a deployed node classifier and observes softmax vectors $\hat{p}_v$ (and, for evaluation, true labels y_v used to build score features, following standard MIA methodology [2], [5]). The adversary’s goal is to decide whether $v \in \mathcal{T}$ or $v \in \mathcal{S}$. Validation nodes are excluded from the final membership evaluation to avoid contamination with early-stopping / defense proxies.*

Equivalently, the adversary plays a membership game: a challenger samples v uniformly from $\mathcal{T} \cup \mathcal{S}$, reveals $(\hat{p}_v, y_v)$, and the adversary outputs a bit $\hat{m} \in \{0, 1\}$. Advantage is $\Pr[\hat{m} = \mathbb{1}[v \in \mathcal{T}]] - \frac{1}{2}$, estimated in aggregate by AUROC over many queries. We emphasize AUROC (and TPR at low FPR) rather than balanced accuracy alone, following Carlini et al. [4].

The adversary has no access to gradients, hidden embeddings, or training randomness beyond what is revealed by posteriors. White-box, link-stealing, and structure-manipulation attacks are out of scope [7].

a) Threat-model boundary (partial knowledge).: An adversary who knows LTE is used could try to reverse-engineer

high-risk neighborhoods from public topology and concentrate queries there; an adversary with black-box access to known non-members can calibrate ϕ thresholds; and because risk-scaled Laplace release is randomized, a multi-query adversary may average K independent releases to reduce noise. We evaluate defense-aware shadows (same training defense and release API), multi-query averaging ($K \in \{1, 5, 20\}$), and a nonlinear MLP- ϕ attacker. We do *not* claim robustness to white-box gradients, adaptive poisoning of the graph, or unbounded query budgets without a session-level noise/query policy.

C. Attack feature map

Following Salem-style score attacks [2], define

$$\phi_1(v) = \max_c \hat{p}_{v,c}, \quad \phi_2(v) = \hat{p}_{v,y_v}, \quad (1)$$

$$\phi_3(v) = -\sum_{c=1}^C \hat{p}_{v,c} \log(\hat{p}_{v,c} + \varepsilon), \quad \phi_4(v) = -(1 - \phi_1) \log(\phi_1 + \varepsilon). \quad (2)$$

Intuitively, ϕ_1 and ϕ_2 capture confidence, ϕ_3 entropy, and ϕ_4 a modified-entropy margin. These are exactly the features our defense aligns, so the training objective matches the attack surface.

D. Metrics

Utility (higher better): test accuracy, macro-F1, macro AUROC. **Privacy** (lower attack success better): confidence-attack AUROC (logistic regression on ϕ), threshold AUROC on ϕ_2 , shadow-model AUROC, and **LiRA AUROC** with TPR at FPR $\in \{0.001, 0.01\}$ [4]. We also report ECE [15] because overconfidence is a common membership cue. An attack AUROC of 0.5 is chance; values near 1.0 indicate severe leakage.

Remark 1. *Because the setting is transductive, non-members are still present in the graph during training. This is slightly weaker than inductive isolation of entire nodes, but remains the dominant practical semi-supervised GNN deployment mode and still admits nontrivial membership distinguishability through the supervised loss asymmetry.*

IV. SUPPORTING EVIDENCE: STRUCTURE PREDICTS SCORE MIA

Prior work already links structure to GNN membership risk [5], [32]. We do *not* claim a new law. As supporting evidence for interpreting SPAB cells, a ridge fit on a controlled (h, ρ, SNR, M) synthetic grid yields leave-one-regime-out Spearman ≈ 0.70 (MAE ≈ 0.038) for confidence-attack AUROC (Fig. 1). Real graphs separate qualitatively: PubMed near chance; Actor high conf/LiRA; Planetoid shows MLP > GNN on conf AUROC (feature channel), which the SNR axis reproduces inside the same design. Coefficients are modest; we treat this as an *interpretive synthetic grid* for audits, not a contribution equal to SPAB.

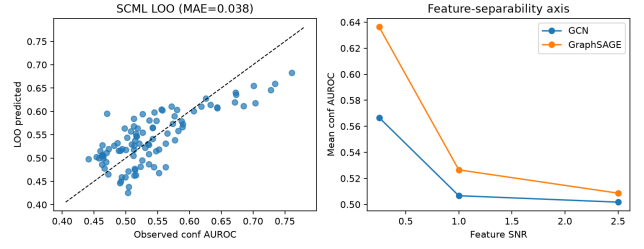


Fig. 1. Supporting LOO plot (not a formal theory): predicted vs. observed conf AUROC on held-out synthetic regimes.

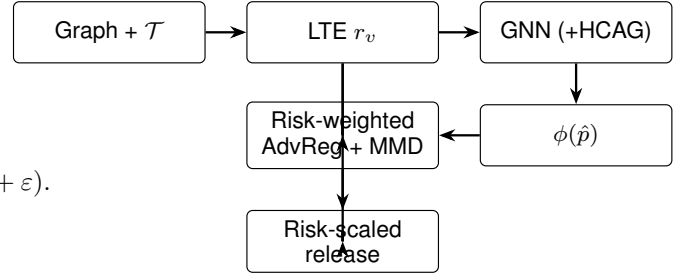


Fig. 2. SAMI: LTE heuristic risk $\rightarrow$ risk-weighted ϕ -alignment; optional release.

V. SAMI: RISK-WEIGHTED ϕ -ALIGNMENT (ONE DEFENSE)

A. Design principle

SAMI estimates per-node structural risk r_v with a three-term heuristic (LTE: inverse degree, local heterophily, supervised-neighbor fraction; equal weight, min-max), then applies risk-weighted ϕ -indistinguishability. HCAG and risk-scaled Laplace release are optional. LTE is a *proxy*, not a learned leakage estimator; we report Spearman(r_v , $|\phi_2\text{-gap}|$) across datasets (Table XV). Removing LTE (uniform weights) collapses Acc (≈ 0.11 on Cora) while privacy barely moves—validity for utility, not proof that LTE is the optimal privacy targeting.

B. Module 1: Leakage Topology Estimator (LTE)

For each node v , compute three non-negative signals that use only $G, y|_{\mathcal{T}}$, and the train mask (never test labels):

$$s_v^{\text{deg}} = \frac{1}{1 + \deg(v)}, \quad (3)$$

$$s_v^{\text{het}} = 1 - \hat{h}_v, \quad \hat{h}_v = \frac{\sum_{u \in \mathcal{N}(v) \cap \mathcal{T}} \mathbb{1}[y_u = y_v]}{\max(1, |\mathcal{N}(v) \cap \mathcal{T}|)}, \quad (4)$$

$$s_v^{\text{sup}} = \frac{|\mathcal{N}(v) \cap \mathcal{T}|}{\max(1, \deg(v))}. \quad (5)$$

Nodes with no train neighbors receive mid-scale heterophily risk (0.5) rather than a pathological maximum. Define the raw score as the mean of (3)–(5) and min-max normalize to $r_v \in [0, 1]$. **Ablation** —LTE replaces r_v with the constant 1 (uniform weighting).

Algorithm 1 SAMI training and release (single trial)

Require: Graph G , masks $\mathcal{T}, \mathcal{V}, \mathcal{S}$, model f_θ , λ, η, σ , epochs T

- 1: Compute LTE risks $\{r_v\}$ via (3)–(5); normalize to $[0, 1]$
- 2: **for** epoch = 1 to T **do**
- 3: Forward f_θ (with HCAG gates if enabled)
- 4: Update D on detached ϕ to classify $\mathcal{T}$ vs $\mathcal{V}$
- 5: Update θ by minimizing (6)
- 6: **end for**
- 7: Release $\tilde{p}_v \leftarrow \text{renorm}(\text{softmax}(z_v/T_v) + \text{Lap}(0, \sigma r_v))$
- 8: Evaluate MIAs using members $\mathcal{T}$ and non-members $\mathcal{S}$ only

C. Module 2: Risk-weighted ϕ -indistinguishability

Let D be a small MLP discriminator on $\phi \in \mathbb{R}^4$. Each epoch: (i) update D to separate $\mathcal{T}$ from $\mathcal{V}$ on *detached* ϕ ; (ii) update the GNN to minimize

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\mathcal{T}) + \lambda \mathbb{E}_{v \sim \mathcal{T} \cup \mathcal{V}} [r_v \text{BCE}(D(\phi(v)), 0)] + \lambda \text{MMD}_r(\phi_{\mathcal{T}}, \phi_{\mathcal{V}}) + \eta \mathbb{E}_{v \in \mathcal{T}} [r_v (-\text{H}(\hat{p}_v))], \quad (6)$$

where the AdvReg term pushes all scored nodes toward the non-member decision of D , MMD_r matches risk-weighted first and second moments of ϕ between train and validation, and the entropy term discourages overconfidence on high-risk supervised nodes. **Critical hygiene:** validation—not test—is the non-member proxy, so the final $\mathcal{T}$ vs. $\mathcal{S}$ evaluation is uncontaminated.

D. Module 3: Homophily-Conditioned Aggregation Gate (HCAG)

For gated GCN/SAGE variants, a per-edge gate

$$g_{uv} = \sigma(\text{MLP}([h_u \| h_v \| r_u \| r_v])) \quad (7)$$

reweights second-layer aggregation. Gates are initialized near 1 (open) for stable early training. HCAG is the architectural counterpart to LTE: it softens message paths likely to create train/test posterior asymmetry under heterophily.

E. Module 4: Risk-scaled release

At inference we optionally apply temperature $T_v = 1 + \beta r_v$ and add Laplace noise with per-node scale σr_v , then clip and renormalize. Full SAMI uses locked $\sigma=0.35$ (Table IX). The identical API transform is applied to shadow models so attackers face the defended interface.

F. Complexity and systems cost

LTE is a single sparse pass over edges: $O(|E|)$. HCAG adds a small MLP evaluation per edge in the gated layer: $O(|E|d_h)$. The discriminator operates on $O(n)$ four-dimensional vectors and is negligible beside GNN training. Risk-scaled release is $O(nC)$ and matches the cost tier of LBP. Shadow/LiRA evaluation multiplies training cost by n_{shadows} and should be reported separately from defense overhead in deployment audits. On PubMed GraphSAGE, wall-clock training rises from ≈ 7 s (none) to ≈ 22 s (full SAMI with HCAG) on CPU

TABLE IV
SYSTEMS OVERHEAD ON PUBMED GRAPH SAGE (CPU, SEED 42, NO LiRA SHADOWS).

Defense	Wall (s)	Train (s)	Peak MB
none	6.9	6.4	5.9
SAMI (–HCAG)	13.9	13.4	6.2
SAMI (full)	22.1	21.6	6.2

TABLE V
DEFENSE HYPERPARAMETERS (FIXED ACROSS CONFIRMATORY RUNS).

Defense	Setting
DropEdge	$r_{\text{de}} = 0.3$
Label smoothing	$\alpha = 0.1$
Early stopping	patience 15, $\delta = 10^{-4}$
Confidence masking	top- $k = 2$
Edge sparsification	$r_{\text{es}} = 0.2$
LBP	Laplace scale 0.3
GTD-style	$\gamma = 1.0$, stage-1 fraction 0.5, pseudo-conf 0.8 (citation); Volume-s
SAMI	$\lambda = 0.5$, $\sigma = 0.35$, warmup 5, HCAG on (locked)

(Table IV). **ogbn-arxiv** (~ 169 k nodes) is enabled via NeighborLoader (HCAG off at Volume); we report Acc/conf/LiRA, train seconds, peak RSS, and defended API QPS in Section VII and Table XI. LiRA shadow counts are *protocol-sized by wall-clock*, not attack-tuned: confirmatory citation cells use $n_{\text{shadows}}=4$; Volume ogbn uses $n_{\text{shadows}}=2$ from the smoke gate in `results/ogbn_smoke_timing.json` (one-shadow wall-clock projection under a ~ 12 – 18 h CPU budget). We do not claim a camera-ready “target 8” unless that run is executed.

G. What SAMI is not

SAMI is not a substitute for DP-SGD when a cryptographic or legal privacy budget is required. It is a structure-aware hardening of the prediction API against score-based membership adversaries, designed to be drop-in compatible with standard GCN/SAGE training loops and YAML-driven experiment grids.

VI. EXPERIMENTAL PROTOCOL**A. Pre-declared evaluation rules**

We freeze the protocol before interpreting confirmatory results [4]: primary privacy = LiRA AUROC (+ TPR@FPR); primary utility = test accuracy / macro-F1 / macro AUROC; secondary = confidence/threshold/shadow AUROC and ECE. Defense hyperparameters are **not** selected to minimize *test* attack AUROC.

B. Fair training

All methods share Adam settings, epoch budget, and architecture width unless a baseline’s definition requires a specific mechanism (e.g., DropEdge rate 0.3, LBP scale 0.3, GTD flattening weight $\gamma=1$). SAMI uses the locked config $\lambda=0.5$, warmup 5, $\sigma=0.35$ (Table IX). Table V lists defense hyperparameters held fixed across the confirmatory grid.

C. Baselines

none; DropEdge; label smoothing ($\alpha=0.1$); early stopping; confidence masking (top-2); MaskArmor-style top-1 masking; edge sparsification; **LBP**; **GTD-style** alternate+flattening; structure-blind **AdvReg**; and a DP-SGD / DP-GNN *reference point* on the Pareto plot (formal privacy, typically steep utility cost)—not claimed as a SAMI competitor under the same utility budget.

D. Attacks and statistics

Confidence, threshold, shadow, and Gaussian LiRA with $n_{\text{shadows}}=4$ in the confirmatory citation core. Volume (ogbn-arxiv) uses $n_{\text{shadows}}=2$ because a one-shadow smoke gate projected the 4-defense $\times$ 3-seed grid outside the CPU budget at $n=4$; the reduction is a *systems* choice, not a privacy-metric cherry-pick. **Defense-aware shadows**: every shadow applies the same training defense and release transform as the target (including SAMI noise), so confidence/threshold/shadow/LiRA attackers are not evaluated against a mismatched undefended API. We additionally report an MLP- ϕ attacker and multi-query averaging ($K \in \{1, 5, 20\}$) against randomized release. Five seeds $\{42, 123, 456, 789, 1024\}$ on citation/Actor cells; three seeds on Volume; paired t -tests vs. none; Holm–Bonferroni adjustment within the confirmatory family; 95% bootstrap CIs over seeds *and* bootstrap CIs on the paired effect ΔAUROC itself; effect sizes ΔAUROC and $\Delta\text{accuracy}$.

a) *Power analysis.*: On Cora GraphSAGE with locked $\sigma=0.35$ and $n_{\text{shadows}}=8$ (3 seeds), LiRA ΔAUROC (SAMI–none) has mean -0.153 with paired SD 0.016; we treat this as a *sensitivity check* (not a fully powered multi-seed LiRA study) and still report five-seed means elsewhere. Conf ΔAUROC remains secondary.

E. Ablations

Full SAMI; $-\text{LTE}$; $-\text{Adv}/\text{MMD}$ ($\lambda=0$ with release noise); $-\text{HCAG}$; temperature-only; AdvReg.

F. Negative controls

We include (i) regimes where leakage is already near chance (PubMed; high-homophily dense synthetics), where large AUROC drops would be spurious; (ii) comparison against a random-guess floor of 0.5; and (iii) reporting of train–test accuracy gaps alongside attack AUROC to connect overfitting diagnostics [3] with membership risk.

VII. RESULTS

Unless noted, citation numbers are five-seed means from `results/core_results.csv`; Volume/Actor/high-risk cells use the seeds stated in each table. The public SPAB report (`results/spab_report.csv`) matches these rows column-for-column.

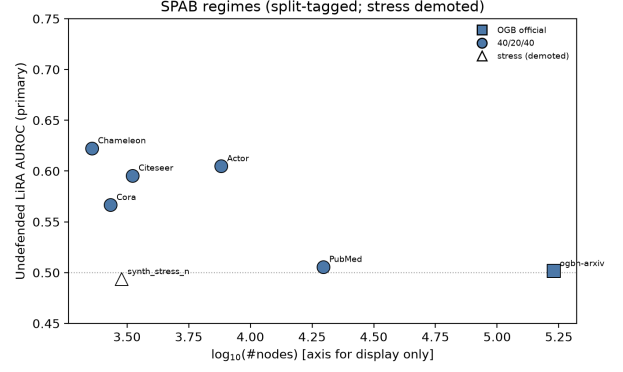


Fig. 3. SPAB regime sketch (undefended *LiRA* AUROC where available, else conf): markers tagged by split protocol. Points are not commensurate on a single causal n -axis—ogbn uses the official OGB split; citation/Actor use 40/20/40; the synthetic stress cell is demoted (open marker).

TABLE VI
CITESEER GRAPH SAGE (5 SEEDS; $n_{\text{shadows}}=4$). SAMI-SEL = VAL-ACC SELECTED; SAMI-LOCK = CORA LOCKED CONFIG.

Defense	Acc $\uparrow$	LiRA $\downarrow$	Conf $\downarrow$	TPR@1% $\downarrow$
none	0.740	0.596	0.741	0.045
GTD-style	0.739	0.551	0.629	0.022
LBP	0.621	0.530	0.629	0.028
MaskArmor	0.740	0.660	0.634	0.000
SAMI-lock	0.731	0.537	0.642	0.026
SAMI-sel	0.744	0.538	0.649	0.023

A. Flagship cell: Cora GraphSAGE (LiRA primary)

Table VII and locked config Table IX. **Primary**: on Cora GraphSAGE, SAMI improves LiRA $0.567 \rightarrow 0.510$ and TPR@1%FPR $0.033 \rightarrow 0.003$ vs. none, beating GTD (LiRA 0.538) and LBP (LiRA 0.645 at Acc 0.701); MaskArmor matches conf but inflates LiRA to 0.580. Conf AUROC is secondary ($0.650 \rightarrow 0.563$). **Scope (own it)**: clean multi-metric wins are Cora GraphSAGE (flagship) and Citeseer GraphSAGE under val-only selected SAMI (Table VI); Chameleon is a third real LiRA win at matched Acc (Table XIV). Cora GCN LiRA change remains tiny ($0.539 \rightarrow 0.533$). **Hard cell (up front)**: `synthetic_low_sparse` GCN fails our success bar (Table VIII). Label smoothing increases conf AUROC on Cora GraphSAGE ($0.650 \rightarrow 0.712$).

B. Citeseer GraphSAGE (val-only second cell)

Pre-registered rule: select $(\lambda, \sigma) \in \{0.5, 1.0\} \times \{0.25, 0.35\}$ by mean *val* Acc on seeds $\{42, 123\}$ only; confirm 5 seeds. Selected: $\lambda=1.0$, $\sigma=0.25$ (Table VI). LiRA $0.596 \rightarrow 0.538$ ($\Delta=-0.058$) with Acc $0.740 \rightarrow 0.744$ vs. none—second flagship bar met. Locked Cora config ($\lambda=0.5$, $\sigma=0.35$) also drops LiRA to 0.537 at Acc 0.731 (Acc within 0.01). LBP is lower on LiRA (0.530) at Acc 0.621; MaskArmor inflates LiRA to 0.660.

A Cora GraphSAGE check with $n_{\text{shadows}}=8$ (3 seeds) preserves—and strengthens—the ordering: none LiRA $0.704 \rightarrow$ SAMI 0.551 at Acc $0.879 \rightarrow 0.873$; confirmatory

TABLE VII

MAIN RESULTS (MEAN OVER 5 SEEDS UNLESS NOTED). ACC = TEST ACCURACY; CONF/LIRA = ATTACK AUROC; TPR@1% = LIRA TPR AT 1% FPR.

Dataset	Defense	GraphSAGE				GCN			
		Acc $\uparrow$	Conf $\downarrow$	LiRA $\downarrow$	TPR@1% $\downarrow$	Acc $\uparrow$	Conf $\downarrow$	LiRA $\downarrow$	TPR@1% $\downarrow$
Cora	none	0.877	0.650	0.567	0.033	0.865	0.581	0.539	—
Cora	label smoothing	0.879	0.712	0.591	0.038	0.868	0.616	0.549	—
Cora	DropEdge	0.874	0.680	0.575	—	0.869	0.573	0.537	—
Cora	GTD-style	0.876	0.585	0.538	0.014	0.874	0.565	0.535	—
Cora	LBP	0.701	0.579	0.645	0.021	0.683	0.562	0.562	—
Cora	MaskArmor	0.877	0.562	0.580	0.000	—	—	—	—
Cora	SAMI	0.874	0.563	0.510	0.003	0.862	0.567	0.533	—
Citeseer	none	0.740	0.741	0.596	—	0.717	0.641	0.560	—
Citeseer	GTD-style	0.739	0.629	0.551	—	0.724	0.653	0.546	—
Citeseer	LBP	0.613	0.656	0.696	—	0.596	0.627	0.593	—
Citeseer	SAMI	0.745	0.657	0.590	—	0.724	0.633	0.575	—

TABLE VIII

GCN HARD CELL ON SYNTHETIC_LOW_SPARSE (5-SEED CONFIRM; ARCH-AWARE LTE). SUCCESS BAR: $\Delta\text{CONF}/\text{LiRA} \leq -0.03$ WITH $\Delta\text{ACC} \geq -0.03$ — NOT MET.

Defense	Acc	Conf	LiRA
none	0.773	0.587	0.502
SAMI (best)	0.746	0.593	0.499

TABLE IX

LOCKED SAMI HYPERPARAMETERS. CORA/ACTOR/CHAMELEON USE THE ROW BELOW; CITESEER USES VAL-ACC SELECTED $(\lambda, \sigma) = (1.0, 0.25)$ (PRE-REGISTERED; TABLE VI). NO TEST-ATTACK TUNING.

Knob	Value
λ , warmup, σ	0.5, 5, 0.35
LTE / arch-aware / HCAG	on / on / on (citation); HCAG off at ogbn
entropy coef, budget B	0.05, 0

TABLE X

ABLATIONS ON CORA (MEANS, 5 SEEDS).

Model	Variant	Acc	Conf
GraphSAGE	full SAMI	0.881	0.574
GraphSAGE	—LTE	0.772	0.571
GraphSAGE	none	0.877	0.650
GCN	full SAMI	0.862	0.567
GCN	—LTE	0.752	0.561

tables use $n=4$ (citation) and $n=2$ (ogbn smoke wall-clock), never attack-tuned.

C. Intervention validity (—LTE)

Fig. 4 / Table X: removing LTE collapses Cora Acc by ≈ 0.11 while conf AUROC barely moves (0.574 vs. 0.571). So —LTE validates that risk weighting is needed for *utility under alignment*, not that LTE is a calibrated privacy targeting mechanism. HCAG and release noise remain optional knobs.

D. ogbn-arxiv: systems + negative control (not a threat proof)

Table XI: NeighborLoader GraphSAGE on $\sim 169k$ nodes. Under the official OGB split, LiRA/conf sit near chance

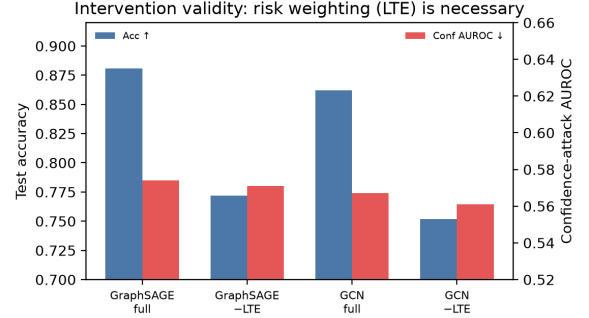


Fig. 4. Full SAMI vs. —LTE on Cora: Acc collapses without risk weighting.

($n_{\text{shadows}}=2$ in the main systems grid). Table XIX: none+SAMI at $n_{\text{shadows}}=4 \times 3$ seeds remains LiRA ≈ 0.503 —a credibility check that the negative control is not an artifact of underpowered shadows. This validates that the audit *scales* and that *this protocol finds little score MIA here*; it does not prove large graphs are safe in general, nor that “Volume = leakage.” Volume-safe GTD (stage-1 CE) matches undefended Acc after an unstable NeighborLoader pseudo stage collapsed Acc—we report the fair CE peer, not the collapsed artifact. LBP taxes Acc without buying LiRA.

a) *Controlled stress test (not a Volume peer)*.: Table XII is a synthetic stress cell ($n=3k$, $\text{Acc} \approx 0.29$, $\text{gen-gap} \approx 0.41$): conf AUROC is high while **LiRA stays at chance**. We treat it as a controlled overfitting stress test for conf-based scores—not evidence that Volume graphs leak, and not commensurate with ogbn on the regime map. Likely $\text{conf} \gg \text{LiRA}$ causes: diffuse posteriors under failed Acc, shadow-likelihood noise, and membership signal concentrated in max-confidence features (see `lte_mechanism_audit.json`).

On Actor, Acc is low for *all* methods (≈ 0.29 – 0.34). Honest LiRA ranking: LBP (0.543) < SAMI (0.552) < GTD (0.585) < none (0.605) $\ll$ MaskArmor (0.797). Thus SAMI improves over none/GTD/MaskArmor on LiRA; LBP is stronger on LiRA alone at worse Acc (0.289 vs. 0.317). Val-only $\sigma \in \{0.25, 0.35\}$ selection on Actor re-selects $\sigma=0.35$ (Ta-

TABLE XI
OGBN-ARXIV GRAPH SAGE (3 SEEDS; $n_{\text{shadows}}=2$). SYSTEMS + PRIVACY.

Defense	Acc $\uparrow$	Conf $\downarrow$	LiRA $\downarrow$	Train s
none	0.666	0.527	0.502	96
SAMI	0.643	0.528	0.499	108
GTD-style	0.664	0.526	0.501	197
LBP	0.454	0.508	0.499	104

TABLE XII
CONTROLLED STRESS TEST (NOT VOLUME PEER): GCN $n=3\text{k}$, LOW- h /LOW-SNR; ACC COLLAPSES; LiRA $\approx$ CHANCE WHILE CONF AUROC IS HIGH.

Defense	Acc $\uparrow$	Conf $\downarrow$	LiRA $\downarrow$	Gen. gap
none	0.291	0.794	0.494	0.414
SAMI	0.261	0.682	0.502	0.242

TABLE XIII
ACTOR GRAPH SAGE BASELINES (5 SEEDS; $n_{\text{shadows}}=4$; LOCKED SAMI). ACC IS LOW ON THIS HETEROPHILIC GRAPH FOR ALL METHODS.

Defense	Acc $\uparrow$	LiRA $\downarrow$	Conf $\downarrow$	Train s
none	0.335	0.605	0.875	3.5
GTD-style	0.339	0.585	0.847	2.9
LBP	0.289	0.543	0.737	3.0
MaskArmor	0.335	0.797	0.790	3.1
SAMI (locked)	0.317	0.552	0.774	9.0

TABLE XIV
CHAMELEON GRAPH SAGE BASELINES (5 SEEDS; $n_{\text{shadows}}=4$; LOCKED SAMI $\sigma=0.35$).

Defense	Acc $\uparrow$	LiRA $\downarrow$	Conf $\downarrow$	Train s
none	0.449	0.623	0.880	8.1
GTD-style	0.475	0.581	0.787	8.6
LBP	0.389	0.544	0.761	8.8
MaskArmor	0.449	0.774	0.757	7.5
SAMI (locked)	0.460	0.551	0.789	25.3

ble IX)—no post-hoc Actor retune. We do not claim an Actor SAMI win on the joint (Acc, LiRA) frontier—only a multi-k heterophilic cell with a complete baseline grid.

E. Chameleon: second heterophilic real grid

Table XIV: locked SAMI on Wikipedia Chameleon ($h \approx 0.24$). LiRA $0.623 \rightarrow 0.551$ ($\Delta = -0.071$) with Acc $0.449 \rightarrow 0.460$ vs. none—a second real-data LiRA win at matched-or-better Acc. vs. GTD (LiRA 0.581) and MaskArmor (LiRA 0.774 inflate); LBP remains slightly lower on LiRA (0.544) at much worse Acc (0.389). Together with Cora GraphSAGE, this is the multi-regime privacy story.

F. LTE is a proxy (with quintile attribution)

Table XV: Spearman(r_v , $|\phi_2|$ -gap) is weak-to-modest on Cora/Citeseer and ≈ 0 on Actor—LTE is not a calibrated leakage estimator. Table XVI: nonetheless, under locked SAMI the drop in conf AUROC is concentrated in high-LTE quintiles (Cora Q5 $\Delta = -0.15$ vs. Q1 $\Delta = +0.12$; Actor Q5 $\Delta = -0.15$

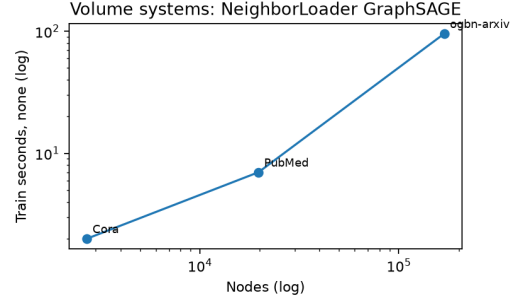


Fig. 5. Train time vs. graph size (ogbn systems).

TABLE XV
LTE VS. $|\phi_2|$ -GAP SPEARMAN (UNDEFENDED GRAPH SAGE).

Dataset	Spearman	p
Cora	0.077	$< 10^{-3}$
Citeseer	0.118	$< 10^{-8}$
Actor	-0.007	0.57

TABLE XVI
CONF AUROC BY LTE RISK QUINTILE (3 SEEDS; GRAPH SAGE). $\Delta = \text{SAMI} - \text{NONE}$.

Dataset	Q1 Δ	Q3 Δ	Q5 Δ	Q5-Q1
Cora	+0.12	-0.10	-0.15	-0.27
Actor	0.00	-0.07	-0.15	-0.15

vs. Q1 ≈ 0). So LTE is a useful *budgeting* heuristic for where alignment spends privacy effort, even when it is a weak absolute predictor of ϕ -gap.

G. How to audit your API (SPAB)

- 1) Fix the membership predicate (supervised mask vs. graph visibility).
- 2) Measure (h, ρ) and choose M ; record n and split protocol.
- 3) Run defense-aware confidence + LiRA (report TPR@1%FPR); log train s / RSS / QPS.
- 4) Emit one SPAB row: $(h, \rho, M, n, \text{Acc}, \text{LiRA}, \text{TPR}, s, \text{MB}, \text{QPS})$.
- 5) If LiRA $\gg 0.5$ and Acc budget allows, apply risk-weighted hardening (SAMI) or DP; re-audit.

Fig. 6 summarizes NeighborLoader audit train cost vs. n (ogbn QPS $\approx 27\text{k}$).

H. Calibration (secondary)

Label smoothing worsens ECE and conf AUROC on Cora GraphSAGE (Table XVII). Label-only gap moves only marginally under SAMI—strong claims target score-based APIs. The synthetic LOO plot (Fig. 1) remains supporting evidence only.

I. Protocol limitations

- **Splits.** Citation/Actor use 40/20/40 random masks (not Planetoid public splits) for seed-wise membership resam-

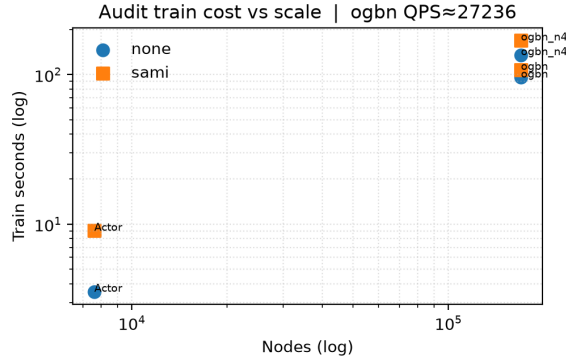


Fig. 6. Audit train cost vs. graph size (none/SAMI). Systems pillar stands even when Volume LiRA is near chance.

TABLE XVII
CORA GRAPH SAGE: CALIBRATION AND PRIVACY.

Defense	Acc	ECE	Conf	LiRA
none	0.877	0.051	0.650	0.567
label smoothing	0.879	0.111	0.712	0.591
LBP	0.701	0.132	0.579	0.645
GTD-style	0.876	0.092	0.585	0.538
SAMI	0.874	0.097	0.563	0.510

TABLE XVIII
PLANETOID PUBLIC-SPLIT APPENDIX: CORA GRAPH SAGE (5 SEEDS;
 $n_{\text{shadows}}=4$). PRIMARY PAPER USES 40/20/40.

Defense	Acc	Conf	LiRA
none	0.788	0.941	0.532
GTD-style	0.821	0.858	0.531
SAMI (locked)	0.837	0.635	0.511

pling; Appendix Table XVIII adds Cora GraphSAGE on the public Planetoid split (none/GTD/SAMI).

- **ogbn LiRA** n_{shadows} . Main Volume grid uses $n=2$ (smoke budget); Table XIX reports none+SAMI at $n=4 \times 3$ seeds for credibility.
- **Metric emphasis**. Protocol primary = LiRA; some plots/stress cells show conf AUROC—we label them secondary.
- **SAMI/val**. Training uses val as the non-member proxy for AdvReg/MMD (never test); this matches defense-aware evaluation but couples the alignment objective to the split.
- **Volume GTD**. We report stage-1-only CE after NeighborLoader pseudo-label collapse; this is a fair Acc peer, not a claim that full two-stage GTD transfers unchanged.
- **No second real Volume graph**. Reddit/ogbn-products were not cached; Chameleon provides a second heterophilic real grid (not Volume-scale).
- **Locked σ** . Paper tables use $\sigma=0.35$ (Table IX) for citation/Actor/Chameleon; ogbn SAMI uses NeighborLoader config ($\sigma=0.1$, HCAG off).

TABLE XIX
OGBN-ARXIV LiRA CREDIBILITY: NONE+SAMI, $n_{\text{shadows}}=4$, 3 SEEDS
(OFFICIAL SPLIT). STILL NEAR CHANCE.

Defense	Acc	Conf	LiRA
none	0.666	0.527	0.503
SAMI	0.643	0.528	0.503

VIII. DISCUSSION

a) *Operators.*: Treat privacy as workload-dependent: emit one SPAB row per (h, ρ, M, n) and spend hardening budget only where LiRA $\gg 0.5$ and Acc allows. NeighborLoader systems numbers (train s, RSS, QPS) belong in the same row as Acc/LiRA so audit cost is auditable.

b) *Open problems.*: A second *Volume-scale* graph with LiRA > 0.55 would still sharpen the large-graph story; Chameleon/Citeseer already provide additional real regimes. Learned (val-only) risk weights that beat LTE on ϕ -gap correlation remain open.

c) *Artifact.*: SPAB v1.0 schema (SPAB_v1_SPEC.md), frozen results/paper_release/, and build_spab_release.py target byte-identical CSV regeneration. Zenodo deposit accompanies the camera-ready.

IX. CONCLUSION

SPAB is a scale-aware conditional audit protocol: ogbn-arxiv contributes NeighborLoader systems and a trustworthy near-chance negative control under this split—not a demonstration that large graphs always leak. Privacy findings are concentrated on Cora GraphSAGE, Citeseer (val-selected SAMI), and Chameleon; Actor shows when LBP dominates LiRA alone; SAMI is a risk-weighted heuristic with locked/pre-registered configs, Acc-critical –LTE, quintile attribution of the conf drop, and an honest GCN hard cell. Artifacts: spab_report.csv, protocol limitations, and frozen release files.

CODE AVAILABILITY

Configuration, defenses, SCML/Volume scripts, and frozen artifacts: <https://github.com/PrivacyAI-Lab/privacygmn>. See REPRODUCE.md and EVALUATION_PROTOCOL.md.

REPRODUCIBILITY CHECKLIST

- **SPAB report**: results/spab_report.csv / .json (matches paper tables).
- **Config**: experiment_config_confirmatory.yaml; Volume: experiment_config_ogbn.yaml.
- **Protocol**: EVALUATION_PROTOCOL.md.
- **Commands**: run_core_tables.py; run_highrisk_volume_synth.py; build_spab_release.py; freeze_paper_release.py.
- **Hardware**: CPU confirmatory runs; LiRA shadow counts protocol-sized by wall-clock.

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